CprE/SE 4910 WEEKLY REPORT 05

10/14/2026 – 10/20/2026

Group number: sdmay27-14

Project title: 3D Plan Visualization: Automated Building Reconstruction and Walkthrough Generator

Client: Buildertrend

Faculty Advisor: Dr. Islam

Team Members/Role:

- ***Nakshatra Gupta - Communication Lead***
- ***Rishiraj Parmar - Organization Lead***
- ***Ryan McNab - Team Meetings Lead***
- ***Daniel Hargrave - Computer Vision Lead***
- ***Jase Beaubien - Graphics Lead***

Weekly Summary

Worked on the prototype and discussed with Dr. Fila about how to scale back the project in a feasible way. Possibly breaking up the project into different smaller milestones is the plan that the groups think should be brought up with the Buildertrend representative.

Past Week Accomplishments

- Group Work:
    - The group discussed different ways in splitting up the projects into smaller parts. More will be known when the group meets with their Buildertrend representative.
- Nakshatra Gupta:
    - Developed an updated extraction script that works on all four of the client's sample plans, after the first version worked on only one.
    - Resolved the Microsoft Teams access problem with Buildertrend after emailing them about the blank page that team members got when signing in.
- Rishiraj Parmar:
    - Tested floor relationships, story ordering, story heights, stairs, open-to-below connections, and roof references. The current test completed successfully with no validation errors.
- Ryan McNab:

- Further improved the computer vision of the prototype to become more accurate. Many of the walls that were sticking out are now more clean. Missing balconies are now showing up and the exteriors of the buildings have become pretty solid compared to the documents given to the team. Interior was slightly improved, but still corrupted in many areas. Here are examples of the given blueprints after being run through the algorithm.









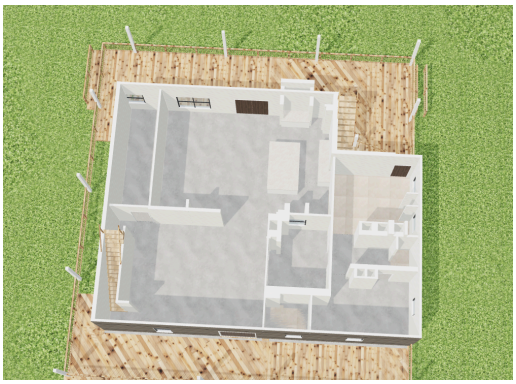
- Daniel Hargrave:
    - Started writing some code for the model. The model is currently taking just the few screenshots that show the floor diagram for the building.
- Jase Beaubien:
    - Planned and researched graphics.
    - Coordinated high-level plans.

Pending issues

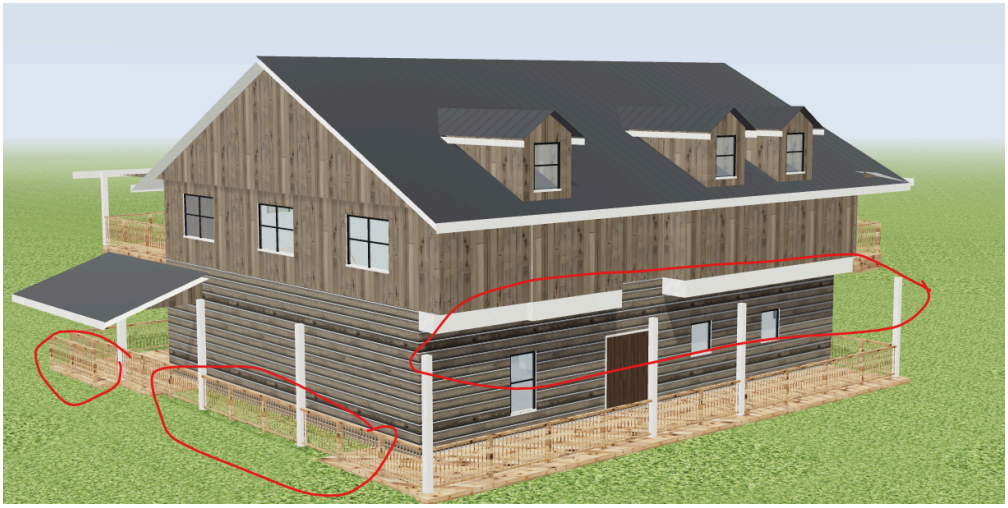
- Nakshatra Gupta:
    - The updated script works on all four sample plans, but it is not known whether future plans from Buildertrend will be formatted differently. More distinct plans may still break it, so it needs testing on any new plans the client sends.
- Rishiraj Parmar: The current prototype relies on already-structured building data and

does not extract information directly from the blueprint. Some values, such as exact story base elevations, also still need to be determined from the plan-processing output before the integration layer can be fully tested.

- Ryan McNab:
    - The algorithm is misreading the furniture as walls and ends up mangling the interior in some places.



    - The roofing and exterior flooring is missing in some places. There are also weird extra railings that need to be cleaned up.



- Daniel Hargrave:
    - It seems unlikely that making a good model with the limited stuff the team has is possible. It might be good to get more stuff from the internet or have an AI create more documents.
- Jase Beaubien:
    - Don't have models to work with.

Individual Contributions

NAME	**Individual Contributions** *(Quick list of contributions. This should*	**Hours this week**	**HOURS cumulative**

	be short.)		
Nakshatra Gupta	Updated extraction script that works on all four sample plans and client communication that resolved the Teams access issue.	3	19
Rishiraj Parmar	Tested and refined the multi-story reconstruction integration prototype using structured data based on the Buildertrend two-story plan.	4	20
Ryan McNab	Once again improved the prototype to be more accurate to the blueprints.	4	25
Daniel Hargrave	Studied the documents provided and tried to separate screenshots into different categories based on what each does. Attempted to come up with some basic plans for how to move forward given the new information.	3	20
Jase Beaubien	Planning and research for graphics.	3	18

Comments and Extended Discussion

The group is questioning the feasibility of the project for what the team has received so far. More will be known in the team's next meeting.

Plans for the upcoming week

- Nakshatra Gupta:
    - Work with the Computer Vision and Graphics teams on how the extracted data fits the shared JSON schema. Test the script on any additional plans the client provides, and follow up with Buildertrend on the open questions.
- Rishiraj Parmar:
    - Continue coordinating the integration format with the shared JSON schema and document any additional fields required by the reconstruction side.

- Ryan McNab:
    - Try to scale down the project with the team and clean up the interior of the blueprints provided by Buildertrend.
- Daniel Hargrave:
    - Finish working on the base model that takes in the basic floor diagrams.
- Jase Beaubien:
    - Further discuss and coordinate the object format for the deserialized plans.
    - Simulate basic models in the program.